

STS 531: Mathematical Statistics I

Homework III

Nolan R. H. Gagnon

May 23, 2026

3.7 Let the number of chocolate chips in a certain type of cookie have a Poisson distribution. We want the probability that a randomly chosen cookie has at least two chocolate chips to be greater than .99. Find the smallest value of the mean of the distribution that ensures this probability.

Solution: Let $X =$ “number of chocolate chips in a certain type of cookie”, with $X \sim \text{Poisson}(\lambda)$. Then

$$P(X \geq 2) = 1 - P(X < 2) \tag{1}$$

$$= 1 - P(X = 1) - P(X = 0) \tag{2}$$

$$= 1 - e^{-\lambda}\lambda - e^{-\lambda} \tag{3}$$

$$= 1 - e^{-\lambda}(\lambda + 1). \tag{4}$$

We require that $P(X \geq 2) > 0.99$, so we must have

$$e^{-\lambda}(\lambda + 1) < 0.01.$$

The smallest λ that satisfies this inequality can be found approximately by numerical methods. Computer software yields $\lambda \approx 6.63835$.

3.8 Two movie theaters compete for the business of 1000 customers. Assume that each customer chooses between the movie theaters independently and with “indifference.” Let N denote the number of seats in each theater. **(a)** Using a binomial model, find an expression for N that will guarantee that the probability of turning away a customer (because of a full house) is less than 0.01. **(b)** Use the normal approximation to get a numerical value for N .

Solution: **(a)** Let $X \sim \text{Binomial}(p = 0.5, n = 1000)$. The probability of turning away a customer is

$$P(X > N) = 1 - P(X \leq N) \quad (5)$$

$$= 1 - \sum_{k=0}^N \binom{1000}{k} (0.5)^k (0.5)^{1000-k} \quad (6)$$

$$= 1 - \sum_{k=0}^N \binom{1000}{k} 0.5^{1000} \quad (7)$$

$$= 1 - 0.5^{1000} \sum_{k=0}^N \binom{1000}{k} \quad (8)$$

Since we want this probability to be less than 0.01, we must have

$$\sum_{k=0}^N \binom{1000}{k} > \frac{0.99}{0.5^{1000}}.$$

(b) Let $Y \sim \text{Normal}(\mu = np = 500, \sigma^2 = np(1 - p) = 250)$. Then

$$P(X > N) = 1 - P(X \leq N) \quad (9)$$

$$\approx 1 - P(Y \leq N) \quad (10)$$

$$= 1 - P\left(\frac{Y - 500}{\sqrt{250}} \leq \frac{N - 500}{\sqrt{250}}\right) \quad (11)$$

$$= 1 - P\left(Z \leq \frac{N - 500}{\sqrt{250}}\right) \quad (12)$$

To satisfy the inequality, we must have

$$P\left(Z \leq \frac{N - 500}{\sqrt{250}}\right) > 0.99.$$

Using a standard normal table, we find that

$$\frac{N - 500}{\sqrt{250}} = 2.33.$$

This gives $N = 537$.

3.13 A truncated discrete distribution is one in which a particular class cannot be observed and is eliminated from the sample space. In particular, if X has range $0, 1, 2, \dots$ and the 0 class cannot be observed (as is usually the case), the 0-truncated random variable X_T has pmf

$$P(X_T = x) = \frac{P(X = x)}{P(X > 0)},$$

with $x = 1, 2, \dots$. Find the pmf, mean, and variance of the 0-truncated random variable starting from **(a)** $X \sim \text{Poisson}(\lambda)$ and **(b)** $X \sim \text{negative binomial}(r, p)$.

Solution: **(a)** The pmf is

$$P(X_T = x) = \frac{e^{-\lambda} \lambda^x}{x!(1 - e^{-\lambda})},$$

where $x = 1, 2, \dots$. The mean is

$$E[X_T] = \sum_{x=1}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!(1 - e^{-\lambda})} \quad (13)$$

$$= \left(\frac{e^{-\lambda}}{1 - e^{-\lambda}} \right) \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} \quad (14)$$

$$= \left(\frac{\lambda e^{-\lambda}}{1 - e^{-\lambda}} \right) \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} \quad (15)$$

$$= \frac{\lambda}{1 - e^{-\lambda}}. \quad (16)$$

The second moment is

$$E[X_T^2] = \left(\frac{e^{-\lambda}}{1 - e^{-\lambda}} \right) \sum_{x=1}^{\infty} \frac{x^2 \lambda^x}{x!} \quad (17)$$

$$= \left(\frac{e^{-\lambda}}{1 - e^{-\lambda}} \right) \sum_{x=1}^{\infty} \frac{x \lambda^x}{(x-1)!} \quad (18)$$

$$= \left(\frac{e^{-\lambda}}{1 - e^{-\lambda}} \right) \left[\sum_{x=2}^{\infty} \left\{ \frac{\lambda^x}{(x-2)!} + \frac{\lambda^x}{(x-1)!} \right\} + \lambda \right] \quad (19)$$

$$= \left(\frac{e^{-\lambda}}{1 - e^{-\lambda}} \right) \left[\lambda^2 e^{\lambda} + \lambda \sum_{x=2}^{\infty} \left[\frac{\lambda^{x-1}}{(x-1)!} \right] + \lambda \right] \quad (20)$$

$$= \left(\frac{e^{-\lambda}}{1 - e^{-\lambda}} \right) [\lambda^2 e^{\lambda} + \lambda e^{\lambda} + \lambda e^{\lambda} + \lambda - \lambda] \quad (21)$$

$$= \frac{\lambda^2 + \lambda}{1 - e^{-\lambda}}. \quad (22)$$

So the variance is

$$\text{Var}[X_T] = \frac{\lambda^2 + \lambda}{1 - e^{-\lambda}} - \frac{\lambda^2}{(1 - e^{-\lambda})^2} = \frac{\lambda(1 - e^{-\lambda} - \lambda e^{-\lambda})}{(1 - e^{-\lambda})^2}.$$

(b) The pmf is

$$P(X_T = x) = \frac{\binom{r+x-1}{x} p^r (1-p)^x}{1-p^r},$$

where $x = 1, 2, \dots$. The mean is

$$E[X_T] = \left(\frac{1}{1-p^r} \right) \sum_{x=1}^{\infty} x \binom{r+x-1}{x} p^r (1-p)^x \quad (23)$$

$$= \frac{r}{1-p^r} \sum_{x=1}^{\infty} \binom{r+x-1}{x-1} p^r (1-p)^x. \quad (24)$$

Let $y = x - 1$. Then the summation becomes

$$\left(\frac{r}{1-p^r} \right) \sum_{y=0}^{\infty} \binom{r+y}{y} p^r (1-p)^{y+1} = \left(\frac{r(1-p)}{p(1-p^r)} \right) \sum_{y=0}^{\infty} \binom{(r+1)+y-1}{y} p^{r+1} (1-p)^y \quad (25)$$

$$= \frac{(1-p)r}{p(1-p^r)}. \quad (26)$$

For a negative binomial random variable X , the 0-th term in all of the moment summations will be 0. Therefore, the variance of X_T can be calculated by simply dividing the variance expression on page 96 of the textbook by $1-p^r$. Thus,

$$Var[X_T] = \frac{r(1-p)}{p^2(1-p^r)}.$$

3.14 Starting from the 0-truncated negative binomial, if we let $r \rightarrow 0$, we get an interesting distribution, the logarithmic series distribution. A random variable X has a logarithmic series distribution with parameter p if

$$P(X = x) = \frac{-(1-p)^x}{x \log p},$$

where $x = 1, 2, \dots$ and $0 < p < 1$. **(a)** Verify that this defines a legitimate probability function. **(b)** Find the mean and variance of X .

Solution: **(a)** Since

$$\sum_{x=1}^{\infty} \frac{-(1-p)^x}{x \log p} = \frac{1}{\log p} \sum_{x=1}^{\infty} \frac{-(1-p)^x}{x} \quad (27)$$

$$= \frac{1}{\log p} \log(1 - (1-p)) \quad (28)$$

$$= \frac{\log p}{\log p} \quad (29)$$

$$= 1, \quad (30)$$

$P(X = x)$ is a legitimate probability function.

(b) The mean of X is

$$E[X] = \sum_{x=1}^{\infty} x \frac{-(1-p)^x}{x \log p} \quad (31)$$

$$= \left(\frac{-1}{\log p} \right) \sum_{x=1}^{\infty} (1-p)^x \quad (32)$$

$$= \left(\frac{-1}{\log p} \right) \left(\frac{1}{1 - (1-p)} - 1 \right) \quad (33)$$

$$= \left(\frac{-1}{\log p} \right) \left(\frac{1}{p} - \frac{p}{p} \right) \quad (34)$$

$$= \left(\frac{-1}{\log p} \right) \left(\frac{1-p}{p} \right) \quad (35)$$

$$= \frac{p-1}{p \log p}. \quad (36)$$

The second moment of X is

$$E[X^2] = \sum_{x=1}^{\infty} x^2 \frac{(1-p)^x}{x \log p} \quad (37)$$

$$= \left(\frac{-1}{\log p} \right) \sum_{x=1}^{\infty} x(1-p)^x \quad (38)$$

$$= \left(\frac{-1}{\log p} \right) \sum_{x=1}^{\infty} xq^x \quad (39)$$

$$= \left(\frac{-1}{\log p} \right) \left(q \frac{d}{dq} \left[\sum_{x=0}^{\infty} q^x - 1 \right] \right) \quad (40)$$

$$= \left(\frac{-1}{\log p} \right) \left(q \frac{1}{(1-q)^2} \right) \quad (41)$$

$$= \frac{p-1}{p^2 \log p}. \quad (42)$$

Therefore, the variance of X is

$$Var[X] = \frac{p-1}{p^2 \log p} - \left(\frac{p-1}{p \log p} \right)^2 \quad (43)$$

$$= \frac{\log(p)(p-1) - (p-1)^2}{p^2 \log^2(p)} \quad (44)$$

$$= \frac{(p-1) [\log(p) - p + 1]}{p^2 \log^2(p)}. \quad (45)$$

3.20 Let the random variable X have the pdf

$$f(x) = \frac{2}{\sqrt{2\pi}} e^{-x^2/2},$$

with $0 < x < \infty$. **(a)** Find the mean and variance of X . **(b)** If X has the folded normal distribution, find the transformation $g(X) = Y$ and values of α and β so that $Y \sim \text{gamma}(\alpha, \beta)$.

Solution: **(a)** The mean of X is

$$E[X] = \int_0^{\infty} x \frac{2}{\sqrt{2\pi}} e^{-x^2/2} dx \quad (46)$$

$$= \frac{-2}{\sqrt{2\pi}} \int_0^{\infty} (-x) e^{-x^2/2} dx \quad (47)$$

$$= \frac{-2}{\sqrt{2\pi}} \left[e^{-x^2/2} \right]_0^{\infty} \quad (48)$$

$$= \frac{2}{\sqrt{2\pi}}. \quad (49)$$

The second moment of X is

$$E[X^2] = \int_0^{\infty} x^2 \frac{2}{\sqrt{2\pi}} e^{-x^2/2} dx \quad (50)$$

$$= \frac{-2}{\sqrt{2\pi}} \int_0^{\infty} (x)(-x e^{-x^2/2}) dx \quad (51)$$

$$= \frac{-2}{\sqrt{2\pi}} \left\{ \left[x e^{-x^2/2} \right]_0^{\infty} - \int_0^{\infty} e^{-x^2/2} dx \right\} \quad (52)$$

$$= \frac{-2}{\sqrt{2\pi}} \left(-\sqrt{\frac{\pi}{2}} \right) \quad (53)$$

$$= 1. \quad (54)$$

Therefore, the variance of X is

$$\text{Var}[X] = 1 - \frac{2}{\pi}.$$

(b) This can be solved using Theorem 2.1.5 from the textbook. We want

$$f_Y(y) = \frac{y^{\alpha-1} e^{-y/\beta}}{\Gamma(\alpha) \beta^{\alpha}}.$$

From theorem 2.1.5, we also know that the nonzero values of f_Y have the form $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$.

Choosing $g^{-1}(y) = \sqrt{y}$ gives

$$f_Y(y) = \frac{2}{\sqrt{2\pi}} e^{-y/2} \frac{1}{2\sqrt{y}} \quad (55)$$

$$= \frac{y^{\frac{1}{2}-1} e^{-y/2}}{\sqrt{2}\sqrt{\pi}} \quad (56)$$

$$= \frac{y^{\frac{1}{2}-1} e^{-y/2}}{2^{\frac{1}{2}} \Gamma\left(\frac{1}{2}\right)} \quad (57)$$

So the transformation is $Y = g(X) = X^2$ and $\alpha = \frac{1}{2}$, $\beta = 2$.